

Rajalakshmi Engineering College

Name: GULZHAR AHAMAD
Email: 240801090@rajalakshmi.edu.in
Roll no: 240801090
Phone: 9043074883
Branch: REC
Department: I ECE FA
Batch: 2028
Degree: B.E - ECE

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 4_MCQ_Updated

Attempt : 1
Total Mark : 20
Marks Obtained : 17

Section 1 : MCQ

1. What does the front pointer in a linked list implementation of a queue contain?

Answer

The address of the last element

Status : Wrong

Marks : 0/1

2. When new data has to be inserted into a stack or queue, but there is no available space. This is known as

Answer

overflow

Status : Correct

Marks : 1/1

3. What is the functionality of the following piece of code?

```
public void function(Object item)
{
    Node temp=new Node(item,trail);
    if(isEmpty())
    {
        head.setNext(temp);
        temp.setNext(trail);
    }
    else
    {
        Node cur=head.getNext();
        while(cur.getNext()!=trail)
        {
            cur=cur.getNext();
        }
        cur.setNext(temp);
    }
    size++;
}
```

Answer

Insert at the rear end of the dequeue

Status : Correct

Marks : 1/1

4. Which one of the following is an application of Queue Data Structure?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

5. What will be the output of the following code?

```
#include <stdio.h>
#include <stdlib.h>
```

```

#define MAX_SIZE 5
typedef struct {
    int* arr;
    int front;
    int rear;
    int size;
} Queue;
Queue* createQueue() {
    Queue* queue = (Queue*)malloc(sizeof(Queue));
    queue->arr = (int*)malloc(MAX_SIZE * sizeof(int));
    queue->front = -1;
    queue->rear = -1;
    queue->size = 0;
    return queue;
}
int isEmpty(Queue* queue) {
    return (queue->size == 0);
}
int main() {
    Queue* queue = createQueue();
    printf("Is the queue empty? %d", isEmpty(queue));
    return 0;
}

```

Answer

Runtime Error

Status : Wrong

Marks : 0/1

6. Which operations are performed when deleting an element from an array-based queue?

Answer

Dequeue

Status : Correct

Marks : 1/1

7. What will be the output of the following code?

```
#include <stdio.h>
#define MAX_SIZE 5
typedef struct {
    int arr[MAX_SIZE];
    int front;
    int rear;
    int size;
} Queue;

void enqueue(Queue* queue, int data) {
    if (queue->size == MAX_SIZE) {
        return;
    }
    queue->rear = (queue->rear + 1) % MAX_SIZE;
    queue->arr[queue->rear] = data;
    queue->size++;
}

int dequeue(Queue* queue) {
    if (queue->size == 0) {
        return -1;
    }
    int data = queue->arr[queue->front];
    queue->front = (queue->front + 1) % MAX_SIZE;
    queue->size--;
    return data;
}

int main() {
    Queue queue;
    queue.front = 0;
    queue.rear = -1;
    queue.size = 0;
    enqueue(&queue, 1);
    enqueue(&queue, 2);
    enqueue(&queue, 3);
    printf("%d ", dequeue(&queue));
    printf("%d ", dequeue(&queue));
    enqueue(&queue, 4);
    enqueue(&queue, 5);
```

```
    printf("%d ", dequeue(&queue));  
    printf("%d ", dequeue(&queue));  
    return 0;  
}
```

Answer

1 2 3 4

Status : Correct

Marks : 1/1

8. In linked list implementation of a queue, the important condition for a queue to be empty is?

Answer

FRONT is null

Status : Correct

Marks : 1/1

9. After performing this set of operations, what does the final list look to contain?

```
InsertFront(10);  
InsertFront(20);  
InsertRear(30);  
DeleteFront();  
InsertRear(40);  
InsertRear(10);  
DeleteRear();  
InsertRear(15);  
display();
```

Answer

10 30 40 15

Status : Correct

Marks : 1/1

10. In a linked list implementation of a queue, front and rear pointers are tracked. Which of these pointers will change during an insertion into a non-

empty queue?

Answer

Only rear pointer

Status : Correct

Marks : 1/1

11. A normal queue, if implemented using an array of size MAX_SIZE, gets full when

Answer

Rear = MAX_SIZE - 1

Status : Correct

Marks : 1/1

12. Which of the following properties is associated with a queue?

Answer

First In First Out

Status : Correct

Marks : 1/1

13. The essential condition that is checked before insertion in a queue is?

Answer

Overflow

Status : Correct

Marks : 1/1

14. In what order will they be removed If the elements "A", "B", "C" and "D" are placed in a queue and are deleted one at a time

Answer

ABCD

Status : Correct

Marks : 1/1

15. The process of accessing data stored in a serial access memory is similar to manipulating data on a

Answer

Queue

Status : Correct

Marks : 1/1

16. Front and rear pointers are tracked in the linked list implementation of a queue. Which of these pointers will change during an insertion into the EMPTY queue?

Answer

Both front and rear pointer

Status : Correct

Marks : 1/1

17. What will the output of the following code?

```
#include <stdio.h>
#include <stdlib.h>
typedef struct {
    int* arr;
    int front;
    int rear;
    int size;
} Queue;
Queue* createQueue() {
    Queue* queue = (Queue*)malloc(sizeof(Queue));
    queue->arr = (int*)malloc(5 * sizeof(int));
    queue->front = 0;
    queue->rear = -1;
    queue->size = 0;
    return queue;
}
int main() {
    Queue* queue = createQueue();
    printf("%d", queue->size);
```

```
    return 0;  
}
```

Answer

0

Status : Correct

Marks : 1/1

18. What are the applications of dequeue?

Answer

All the mentioned options

Status : Correct

Marks : 1/1

19. Which of the following can be used to delete an element from the front end of the queue?

Answer

None of these

Status : Wrong

Marks : 0/1

20. Insertion and deletion operation in the queue is known as

Answer

Enqueue and Dequeue

Status : Correct

Marks : 1/1